Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Sea Level TAC - DUACS products:

SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057

Issue: 3.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/07/27	all	Creation	M. Pujol	
	2018/10/11		Accepted		Mercator Ocean
1.1	2021/01/15	VII	Update of C3S web links	M. Pujol	M. Pujol
2.0	2021/09/10		Update with DT-2021 standards	G. Taburet	M. Pujol
2.1	2021/12/16	II	Change for MY automatic extensions	M-I Pujol	M-I Pujol
3.0	2024/11/06	all	Use new template Upgrade for CMEMS Nov2024 MY & NRT release - Full reprocessing DT-2024	M. Ballarotta	M-I Pujol

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the reprocessed (REP/DT) Data Unification and Altimeter Combination System (DUACS) products in the frame of the Climate Change Copernicus Service (C3S), listed here after. The REP product featured in this document covers the period from 1993 to present. The dataset is regularly updated with a nearly 6-month delay.

Product	SEALEVEL_GLO_PHY_CLIMATE_L4_MY_008_057
Area	Global ocean
Satellites	Merging measurements from two different altimeters with orbits like/close to Topex/Poseidon and ERS-1 historical orbits
Spatial resolution	1/4° x 1/4° Cartesian grid
Temporal resolution	1 day

The number of altimeter data processed by the system varies with time, according to satellite availability. The reader can refer to the C3S Product User Guide Specification (PUGS, see the References section below) document for more details.

I.1.1 CMEMS product and C3S product

Two types of sea level altimetric gridded products generated by the DUACS production system are currently available:

- Global and regional (European seas) gridded products generated and disseminated in the framework of the Copernicus Marine Service (CMEMS).
- Global gridded product generated in the framework of the Copernicus Climate Change (C3S) service, and disseminated both via C3S and CMEMS. No regional product is available for C3S.

This section aims at presenting the particularities of the sea level gridded datasets produced by two distinct approaches. The CMEMS approach focuses on the mesoscale mapping capacity of the altimeter data together with the stability of the overall dataset whereas the C3S products focus on the stability of the global and regional Mean Sea Level (MSL), even if this implies potential reduction of the spatial sampling of the ocean.

The first difference between both datasets is related to the number of altimeters used in the satellite constellation. All available altimeters are included in the CMEMS products whereas a fixed number of altimeters (two) are included in the C3S products. Previous studies (Dibarboure et al., 2011; Pascual et al., 2006) underscored the necessity of a minimum of a two-satellite constellation for the retrieval of mesoscale signals. Within the production process, the long-term stability and large-scale changes are built upon the records from the reference missions (TOPEX-Poseidon, Jason-1, Jason-2 and Jason-3, Sentinel-6A) used in both CMEMS and C3S products. The additional missions (e.g. up to four additional missions in 2017) are homogenized with respect to the reference missions. These additional missions contribute to improve the sampling of mesoscale processes, provide high-latitude coverage and increase the product accuracy. However, the total number of satellites strongly varies during the altimetry era and some biases may appear with the introduction of a new satellite flying on a drifting orbit, which may affect the stability of the global and regional MSL. Even if the spatial sampling is

reduced with less satellites, the risk of introducing such an anomaly is thus reduced in the C3S products and the stability is improved. In the CMEMS products, the stability is ensured by the reference missions. In addition, the mesoscale errors are also reduced in the CMEMS products due to the improved ocean surface sampling thanks to the use of all satellites available in the constellation.

The level 2 altimeter standards used to compute the sea level anomalies in the CMEMS and C3S products are identical. A second difference between these products is related to the reference used to compute the Sea Level Anomalies: either a Mean Sea Surface (MSS) for all missions in the C3S products or a mean profile of sea surface heights used along the theoretical track of the satellites with a repetitive orbit in the CMEMS products. These mean profiles increase the local accuracy of the sea level estimates, but the combined use of both MSS and mean profiles for successive missions in the CMEMS merged products could lead to some biases affecting the sea level stability. Even if this has a minor impact at global scale, the stability of the regional MSL can be affected, particularly in the Mediterranean Sea and Black Sea (not shown here). So, the systematic use of the MSS for all missions contributes to ensure the MSL stability in the C3S products. Furthermore, the accuracy of the CMEMS products is increased with the use of the mean profiles for repetitive missions.

The last difference between these products concerns the temporal and spatial scales used in the mapping process (Optimal Interpolation). It has been reviewed specifically for the CMEMS production and is different in the C3S and CMEMS sea level products.

In conclusion, with the best spatial sampling, the all-satellite CMEMS gridded merged products should be preferred for oceanic mesoscale applications and data-assimilation while the C3S two-satellite gridded merged products should be used for climate applications including mean sea level change, variability and oceanic circulation.

In the future, additional differences may be introduced between these two multi-mission sea level products related to specific processing of some missions. For example, LRM (Low-Resolution Mode) and SAR (Synthetic Aperture Radar) modes of the Jason-CS/Sentinel-6 mission could contribute to differentiate the CMEMS and C3S products.

I.2 Summary of the results

The quality of the **delayed-time C3S sea level products** produced by the DUACS system (**version DT-2024**) has been assessed by comparison with independent measurements (in-situ and satellite) and in coordination with other projects (ESA SL_cci[1], CNES SALP[2] and CMEMS[3]). The results are summarized below.

Sea Level Anomaly (SLA) and Absolute Dynamic Topography (ADT):

The **sea level** long-term **trend** can be monitored with the C3S products. The uncertainty of the mean sea level trend over the global ocean has been estimated to be of the order of 0.4 mm/yr (Ablain et al., 2019). On a regional scale, the averaged local sea level trend uncertainty related to the instrumental observing system only has been estimated to 0.83 mm/yr with values ranging from 0.78 to 1.22 mm/yr (the uncertainty related to the natural ocean variability has to be added; Prandi et al., 2021).

Sea level errors for **mesoscale signals** in the C3S products vary between 1 cm^2 (in low variability areas) to 20.8 cm^2 (in high variability areas). Wavelengths accessible with gridded products are larger than ~200 km.

Geostrophic currents:

Geostrophic currents derived from altimeter gridded products are usually underestimated when compared to the in-situ observations. **Errors** on geostrophic currents have been estimated to range between **11 and 17 cm/s**, depending on the ocean surface variability.

I.3 Estimated Accuracy Numbers

C3S validation results, as well as Estimated Accuracy Numbers for C3S products are presented in the C3S Quality Assessment Report document (see the References section below).

[1] https://climate.esa.int/en/projects/sea-level/
[2] https://www.aviso.altimetry.fr/en/home.html
[3] About Copernicus | CMEMS

II PRODUCTION SYSTEM DESCRIPTION AND VALIDATION RESULTS

DUACS is the system production for C3S products (see also §I.1.1).

A description of this system, and altimeter standards used for the C3S production are available in the C3S PUGS document (see the References section below).

C3S products are generally extended in time (i.e. an additional number of months are added) at least once a year, so that these datasets cover 1993 up to five to twelve months before present. These datasets benefit from the highest quality altimeter measurements and geophysical corrections. In addition, they are produced with a unique system thus no changes in time in the set up are required. They are also extended in time using the same system in order to minimize the risk of quality loss or spurious signals appearing in time.

III VALIDATION FRAMEWORK

III.1 The main sources of errors

The quality of the merged L4 products directly depends on the quality of the L3 products used as input in the L4 processing. Nevertheless, the main source of error comes from the sampling capability of the altimeter constellation. The more altimeters are available, the best is the mesoscale sampling. For C3S products, the number of altimeters is relatively stable over time (mainly two altimeters). Another source of error for L4 products is directly linked to the methodologies and parameters applied for SLA interpolation on a regular grid. Optimal Interpolation used in DUACS processing does not allow the restitution of the full dynamical spectrum, limiting the capability of retrieving small mesoscale in L4 products (Chelton et al., 2011, 2014) .

III.2 Validation methodology

The quality of the global gridded SLA products was estimated by comparison with independent altimeter along-track and tide gauge measurements, with focus on mesoscale signal and coastal signal, respectively. The quality of the geostrophic current was estimated by comparison with independent drifter measurements. The methodologies are fully discussed in Pujol et al. (2016) and Taburet et al. (2019).

IV VALIDATION RESULTS

Here we only summarize the main outcome obtained during the validation of the C3S DT-2024 product version.

IV.1 Variable SLA (Level-4 gridded)

Figure 1 presents the Root Mean Square (RMS) of the difference between the DT-2024 sea level anomaly gridded product and the independent SARAL-DP/ALtiKa along-track measurements over the period from 2018-01-01 to 2018-12-31. It illustrates the inhomogeneity of the SLA gridded product error for the mesoscale signal, particularly between low and high variability areas. Figure 1 compares the mapping error between the DT-2024 and DT-2021 C3S reprocessing. A blue pattern indicates a reduction in mapping error, while a red pattern means an increase of mapping error in the new DT-2024 reprocessing. Overall, the new reprocessing reduces the error. Compared to the previous version of the products (i.e., DT-2021), this error is significantly reduced (up to 8% in coastal areas). The SLA gridded product errors in other areas are estimated as between 2 to 5 cm², with higher values in high variability coastal areas. This error is globally reduced by approximately 1% to 2% compared to the previous version of the products.

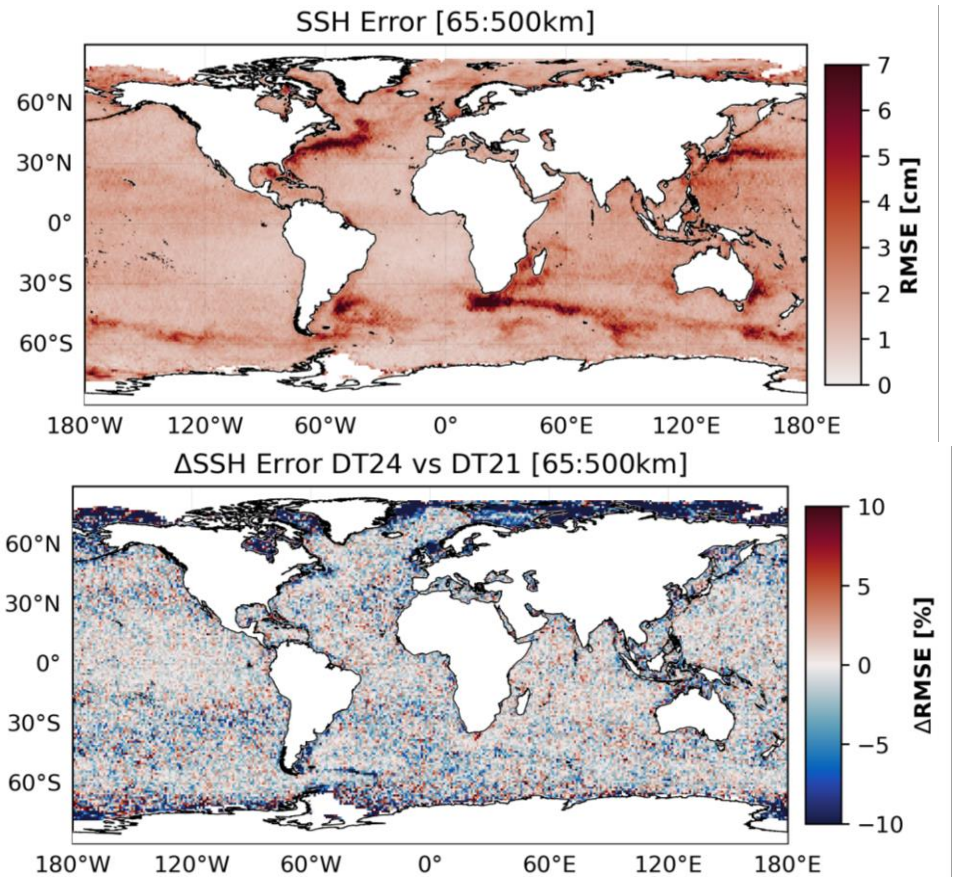

Figure 1: Upper panel: RMS of the difference between global gridded DT-2024 C3S sea level anomaly and independent SARAL-DP/AltiKa along-track measurements over the period 2018-01-01 to 2018-12-31 (units: cm); Lower panel: difference of RMSE between DT-2024 and DT-2021 global gridded products. Negative values (i.e., blue colour) indicates a reduction of error in DT-2024 product.

Tide Gauges (TG) from the GLOSS/CLIVAR network[4] have been used for SLA gridded products validation over the Global Ocean and for the period 1993-2020. Figure 2a presents the RMS of the difference between DT-2024 sea level anomaly gridded product and an independent TG dataset. It also illustrates the inhomogeneity of the SLA gridded product error in coastal areas. The SLA gridded product errors vary between ~1 cm to more than 10 cm. The comparison of altimeter SLA and independent TG measurement shows improved results when the altimeter product DT-2024 version is considered rather than the previous one (DT-2021) (Figure 2b). The Root Mean Square Error (RMSE) between altimetry and TG is especially reduced in part of the European coasts and in the Japanese coast. In those regions, the reduction in the variance of the differences between altimetry and TGs ranges between 0.2% to more than 5% of the TG signal. In some other coastal areas, degradation is however observed. This is the case in the south Pacific Ocean and Indian Ocean. Nevertheless, these degradations are observed for stations where the differences between altimetry and TG reaches less than 1% of the TG signal.

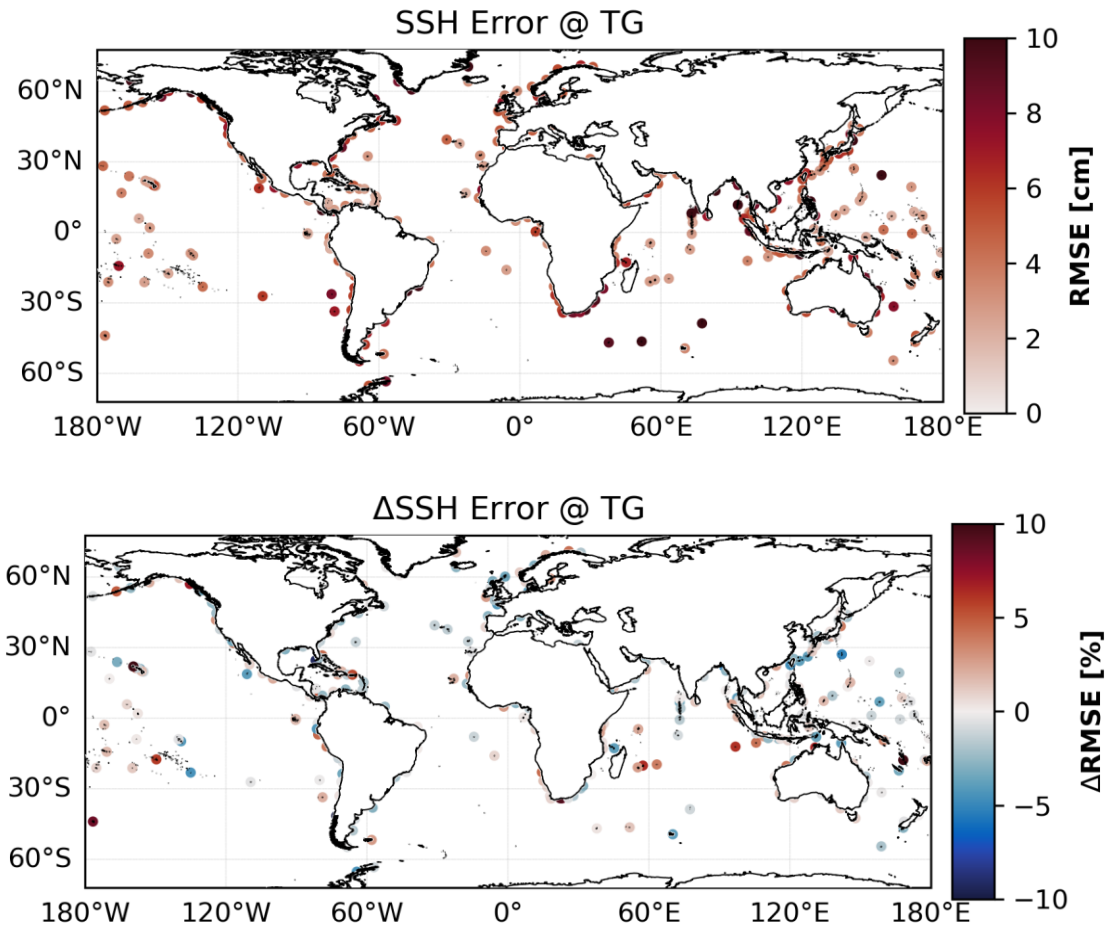

Figure 2: Upper panel: RMS of the difference between global gridded DUACS DT-2024 sea level anomaly and independent tide gauge measurements over the period 1993 to 2020 (units: cm); Lower panel: difference of RMSE between DT-2024 and DT-2021 global C3S gridded products. Negative values (i.e., blue colour) indicates a reduction of error in DT-2024 products.

[4] https://gloss-sealevel.org/.

The errors observed on mesoscales also highlight the L4 product spatial resolution capability. As discussed in Pujol et al. (2016), SLA gridded product effective resolution is constrained by the altimeter sampling capability and mapping methodology used. In order to estimate the spatial resolution of the gridded products, an evaluation has been carried out based on a spectral approach. A full description of this approach can be found in Ballarotta et al. (2019).

Figure 3 (upper panel) shows the global map of the effective spatial resolution of the DT-2024 global maps computed over the period from 2018-01-01 to 2018-12-31, where Saral/AltiKa data were used as an independent dataset. The resulting mean spatial resolution of the global gridded SLA is about 200 km at mid-latitudes. The Observable wavelength increase in the equatorial band to reach ~500 km. The comparison with the spectral content computed from full-resolution Saral/AltiKa 1 Hz along-track measurements (not reported here) shows that a large part of the energy observed in along-track measurements at wavelengths ranging from 200 to 65 km is missing in the SLA gridded products. The loss reaches nearly -40% at wavelength 200 km and rapidly increase at shorter wavelengths. Overall, the DT-2024 SLA gridded C3S products have similar resolution than the previous DT-2021 C3S products.

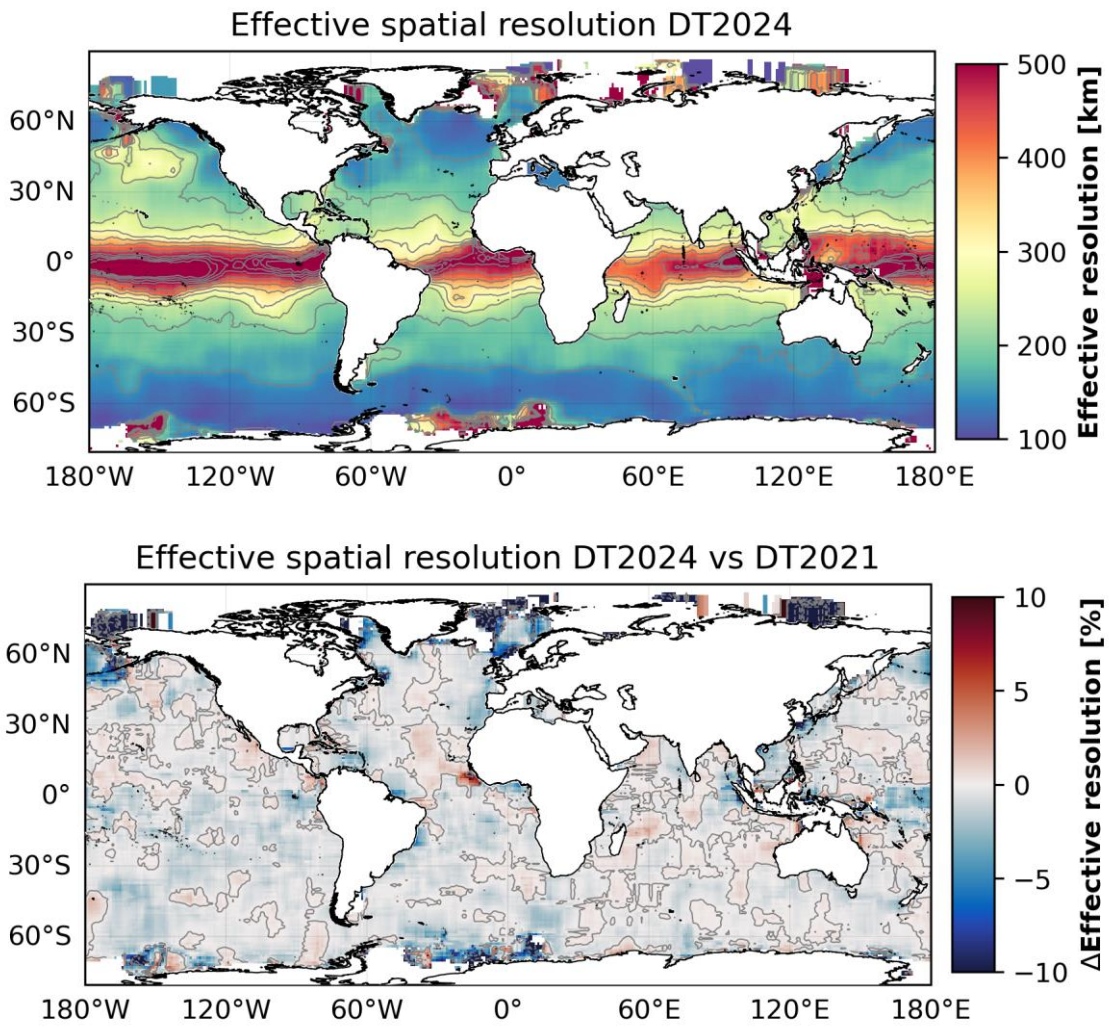

Figure 3: Effective spatial resolution (in km; upper panel) and gain / loss of effective resolution between DT-2024 and DT-2021 products (lower panel).

IV.2 Variable ADT=SLA+MDT

The quality of the ADT field is directly depending on the quality of the SLA and Mean Dynamic Topography (MDT) fields. The SLA error budget is described in the previous chapter. The quality of the MDT fields used in the DUACS processing is described in MDT specific QUID documents for products 008-063, 008-066 and 008-067[5].

IV.3 Variable UV

The absolute surface currents in the product are calculated using the principle of geostrophy from gridded SLA/ADT products. The quality of these products strongly depends on the quality of the SLA/ADT field and on the methodology used to estimate the derivate.

The comparison with drifter measurements gives an indication of the errors on geostrophic current products.

The distribution of the speed of the current (not shown) shows a global underestimation of the current in the altimeter products compared to the drifter observations, especially for currents with medium and strong intensities (> 0.2 m/s). The Figure 4 shows the zonal and meridional RMS differences in 5° x 5° boxes between AOML drifters and absolute geostrophic current products over the period 1993-2020. The equatorial band was excluded from the analysis, since the geostrophic approximation do not lead to an accurate estimation of the currents in this area. Elsewhere, the RMS of the differences is around 11-12cm/s (12.1 cm/s for zonal –11.2cm/s meridional– component). Locally, the RMS of the differences is higher and it reaches 18 cm/s over high variability areas.

The difference with the previous version (DT-2021 standards) of the products is shown in Figure 5 (upper panel). In this Figure, it can be observed that the reduction of the error is mainly found at mid-latitude (5-10%). Larger errors are found for the zonal component at high latitude, especially in the Southern hemisphere.

[5] https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-063-066-067.pdf

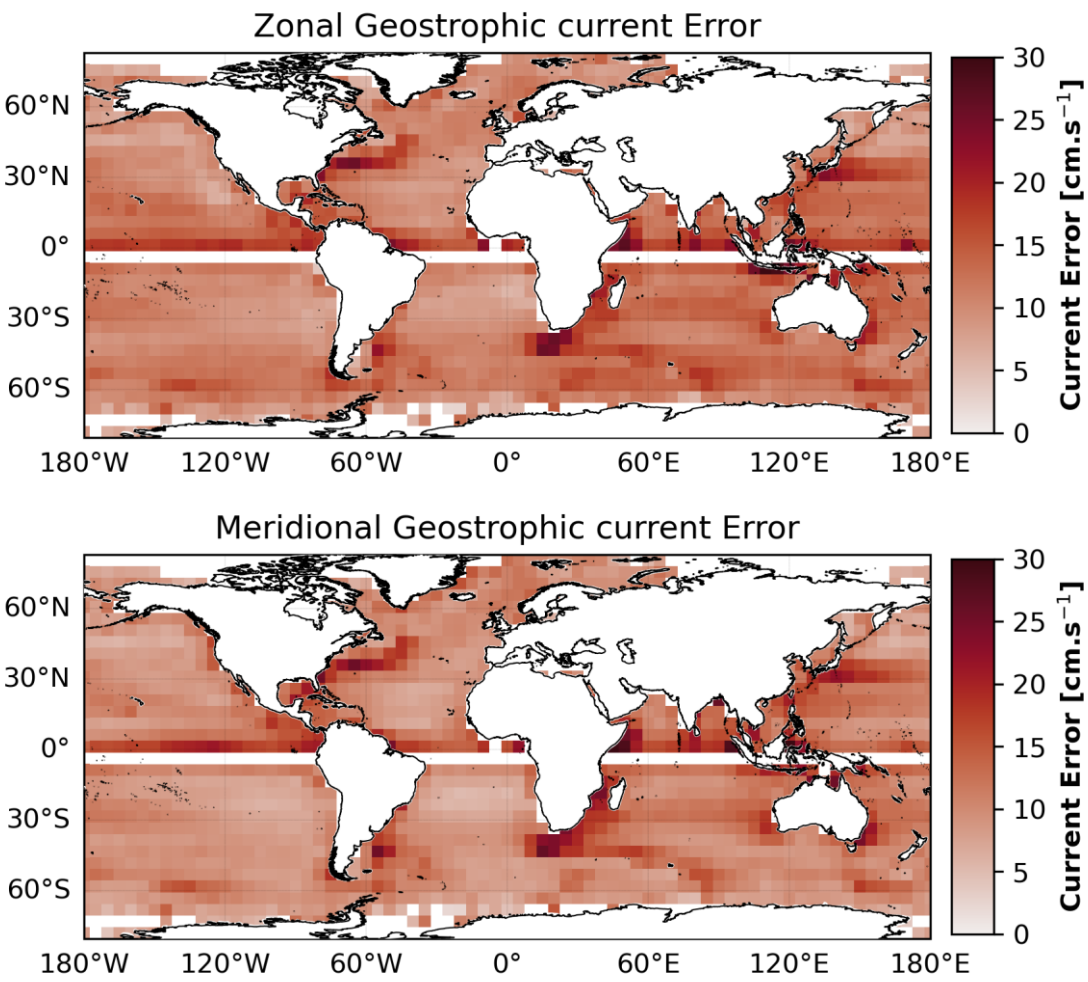

Figure 4: Zonal (upper panel) and meridional (lower panel) RMS of the difference between DT-2024 geostrophic current and drifter's measurements over the period 1993-2019 (units: cm/s).

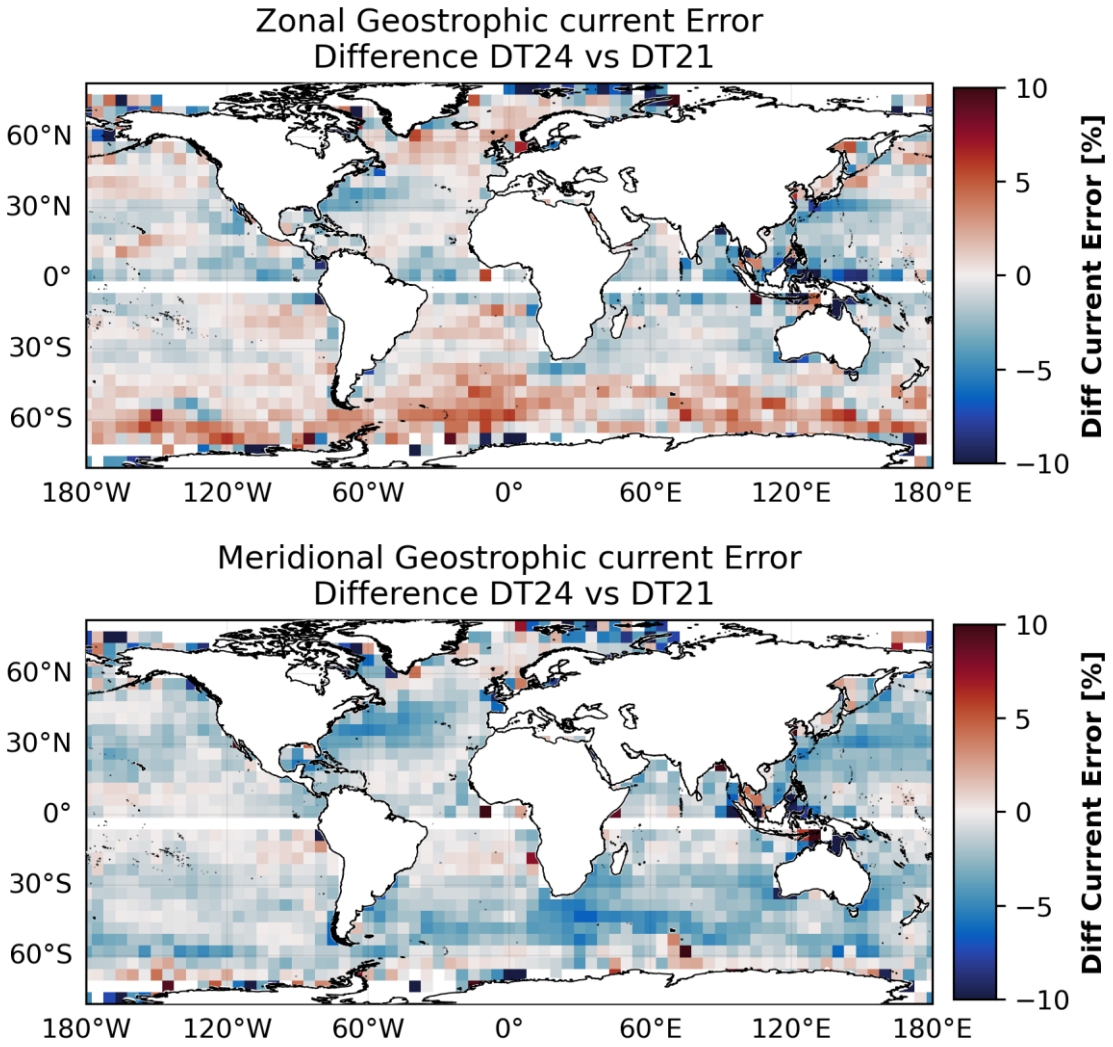

Figure 5: Zonal (upper panel) and meridional (lower panel) difference of RMSE between DT-2024 and DT-2021 gridded geostrophic current products (units: %). Negative values (i.e., blue colour) means a reduction of error in DT-2024 products.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among others, and overall, changes that may impact product quality (Table 1).

Table 1: Overview of the main evolutions implemented in the C3S DT processing.

System version	Product version	Date of change	Description of the change
7.0	*DT-2021*	*Dec 2021*	Full reprocessing taking into account - New altimeter standards and geophysical corrections - New MSS and MP - New mapping parameters - Improved LWE correction - NetCDF format change / product/dataset nomenclature change **DT-2018 version is no longer available**
7.0	*DT-2021*	*Jun 2022*	Temporal extension covering the period 2021/08/03 – 2021/12/31
7.0	*DT-2021*	*Jul 2022*	Temporal extension covering the period 2022/01/01 – 2022/02/09
7.1	*DT-2021*	*Dec 2022*	Temporal extension covering the period 2022/02/10 – 2022/06/23 Sentinel-6A introduced as new reference mission (LR measurements). Jason-3 orbit change; addition of new "j3n" dataset.
7.1	*DT-2021*	*Mar 2022*	Temporal extension covering the period 2022/06/24 – 2022/08/04
7.1	*DT-2021*	*Nov 2023*	Temporal extension covering the period 2022/08/05 – 2023/06/07
7.2	*DT*-2021 interim	*May* 2024	Temporal extension covering the period 2023/06/08 – 2023/09/08
8.0	*DT-2024*	*Dec 2024*	Full reprocessing taking into account - New altimeter standards and geophysical corrections - New MSS - New MDT - Change of the geostrophic current computation method **DT-2021 version is no longer available**

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

The newly reprocessed **C3S climate products** only benefit from updated altimetric standards and level 3 product corrections, while the mapping method remains unchanged. As a result, these new L4 products are generally better than the previous versions, with the most significant improvements seen in coastal regions (up to a 10% reduction in mapping errors), mainly due to enhanced tidal corrections introduced in the new FES2022 tidal solution. Improvements on geostrophic currents are mainly due to the new MDT product used for the C3S products as well as the new method for computing the geostrophic current anomaly (centered finite difference instead of 9-stencil differentiation; Arbic et al., 2012).

VII REFERENCES

C3S PUGS document: available at https://cds.climate.copernicus.eu/datasets/satellite-sea-level-global?tab=documentation

C3S Quality Assessment Report document: available at https://cds.climate.copernicus.eu/datasets/satellite-sea-level-global?tab=documentation

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